

Properties of $\hat{\beta}_{OLS}$

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Outline

- 1 Chapter 4: Properties of $\hat{\beta}$
- 2 Chapter 6: Convergence Concepts
- 3 Chapter 7: Large Sample Properties of $\hat{\beta}$

Assumption 4

So far, the results are algebraic. To make statistical statements about the estimates, we need to make statistical assumptions.

Assumption 4.1 $\{(y_i, x_i)\}$ are i.i.d.

Assumption 4.2

(i) Linear Model: $y_i = x_i' \beta + e_i$ with $\mathbb{E}[e_i | x_i] = 0$

(ii) $\mathbb{E}[y_i^2] < \infty$, $\mathbb{E} \|x_i\|^2 < \infty$, $Q_{XX} = \mathbb{E}[x_i x_i'] > 0$,

Assumption 4.3 $E[e_i^2 | x_i] = \sigma^2(x_i) = \sigma^2$

- No serial correlation and homoskedasticity (spherical errors)

The Conditional Mean Restriction

$$E(e|X) = \begin{pmatrix} E(e_1|X) \\ E(e_2|X) \\ \vdots \\ E(e_n|X) \end{pmatrix} = \begin{pmatrix} E(e_1|x_1, \dots, x_n) \\ E(e_2|x_1, \dots, x_n) \\ \vdots \\ E(e_n|x_1, \dots, x_n) \end{pmatrix} = 0$$

- $E(e_i|X) = 0$, or $E(e_i|x_{j1}, \dots, x_{jK}) \forall i, j = 1, \dots, n$.
 - needed for $E(Y|X) = X\beta$. Correct model specification.
- The iid assumption simplifies $E[e_i|X] = 0$ to $E[e_i|x_i] = 0$.
- $E[x_i e_i] = E_X[x_i E(e_i|x_i)] = 0$, $E(e_i) = E_X[E(e_i|X)] = 0$.

Conditional Variance

Since $E[e|X] = 0$, $\text{var}(e|X) = E(ee'|X)$.

$$E(ee'|X) = \begin{pmatrix} E(e_1 e_1|X) & E(e_1 e_2|X) & \dots & E(e_1 e_n|X) \\ E(e_2 e_1|X) & E(e_2 e_2|X) & \dots & E(e_2 e_n|X) \\ \vdots & & & \\ E(e_n e_1|X) & E(e_n e_2|X) & \dots & E(e_n e_n|X) \end{pmatrix} = \underbrace{D}_{n \times n}$$

- D depends on error structure:
 - D1: Homoskedastic errors: $\text{var}(e_i|X) = \sigma^2 \forall i$
 - D2: Heteroskedastic errors: $\text{var}(e_i|X) = \sigma_i^2$
 - D3: Heteroskedasticity+correlation: $\text{cov}(e_i, e_j|X) \neq 0, i \neq j$

Finite Sample Properties of $\hat{\beta}$

- Conditionally unbiased:

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'y \\ &= (X'X)^{-1}X'(X\beta + e) \\ &= \beta + (X'X)^{-1}X'e \\ E(\hat{\beta}|X) &= \beta + E[(X'X)^{-1}X'e|X] \\ &= \beta + (X'X)^{-1}X'E[e|X] \\ &= \beta.\end{aligned}$$

- Unconditionally unbiased:

$$\begin{aligned}E[\hat{\beta}] &= E_X E(\hat{\beta}|X) \\ &= E_X \hat{\beta} \\ &= \beta.\end{aligned}$$

Conditional Variance: General case

Variance of unbiased estimators with spherical errors:

$$\begin{aligned}
 \hat{\beta} &= \beta + (X'X)^{-1}X'e \\
 \text{var}(\hat{\beta}|X) &= E[(\hat{\beta} - E[\hat{\beta}|X])(\hat{\beta} - E[\hat{\beta}|X])'|X] \\
 &= E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'|X] \\
 &= E[(X'X)^{-1}X'ee'X(X'X)^{-1}|X] \\
 &= (X'X)^{-1}X'E(ee'|X)X(X'X)^{-1} \\
 &= (X'X)^{-1}X'DX(X'X)^{-1} \\
 &\equiv V_{\hat{\beta}} \quad \text{Sandwich Variance}
 \end{aligned}$$

Gauss Markov Theorem

Under homoskedasticity, $\hat{\beta}$ is the best (minimum variance) among all linear unbiased estimators of the form $\tilde{\beta} = A'Y$.

- $\mathbb{E}[\tilde{\beta}|X] = A'\mathbb{E}[Y|X] = A'X\beta$. It is unbiased if $A'X = I_K$.
- OLS takes $A = X(X'X)^{-1}$. Since $A'X = I_K$, it is unbiased.
- Now $\text{var}[\tilde{\beta}|X] = A'\text{var}[Y|X]A = A'DA = A'A\sigma^2$. For OLS to be best, we need to show for other A , $A'A - (X'X)^{-1} \geq 0$.
 - let $C = A - X(X'X)^{-1}$ and note that $X'C = 0$. Then

$$\begin{aligned}
 A'A - (X'X)^{-1} &= (C + X(X'X)^{-1})'(C + X(X'X)^{-1}) - (X'X)^{-1} \\
 &= C'C + C'X(X'X)^{-1} + (X'X)^{-1}X'C \\
 &\quad + (X'X)^{-1}X'X(X'X)^{-1} - (X'X)^{-1} \\
 &= C'C \geq 0.
 \end{aligned}$$

Estimation of σ^2

Recall $\hat{e}_i = y_i - x_i' \hat{\beta} = e_i - x_i'(\hat{\beta} - \beta)$.

- Natural estimator of $\sigma^2 = E[e_i^2]$ is $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \hat{e}_i^2$. But this is a biased estimator
- $\hat{e} = My = Me$ and $\hat{e}'\hat{e} = e'Me$ since M is idempotent

$$\begin{aligned}
 E[\text{trace}(e'Me)|X] &= E[\text{trace}(Mee'|X)] \\
 &= \text{trace}(M\sigma^2 I_n) = \sigma^2 \underbrace{\text{trace}(M)}_{n-K} \\
 &= (n-K)\sigma^2 = E(\hat{e}'\hat{e}|X).
 \end{aligned}$$

- An unbiased estimator of σ^2 is $s^2 = \frac{\hat{e}'\hat{e}}{n-K}$. We estimate $\text{var}(\hat{\beta}|X)$ by $s^2(X'X)^{-1}$.

To see that s^2 is unbiased, consider case $K = 1$.

$$\begin{aligned}s^2 &= \frac{1}{n-1} \sum_i^n (y_i - \bar{y})^2 = \frac{1}{n-1} \sum_i^n [y_i^2 - n\bar{y}^2] \\ E(s^2) &= \frac{1}{n-1} \left[\sum_i E(y_i^2) - nE(\bar{y}^2) \right] \\ &= \frac{1}{n-1} \left[n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) \right] = \sigma^2\end{aligned}$$

where we have used $E(y_i^2) = \sigma^2 + \mu^2$ and $E(\bar{y}^2) = \frac{\sigma^2}{n} + \mu^2$.

In-Sample vs Out-of-Sample Forecast Errors

- We estimate β using data $\{(y_i, x_i)\}_{i=1}^n$ and get the in-sample fit $\hat{y}_i = x_i' \hat{\beta}$, $i = 1, \dots, n$.
- Now we want a 'out-of-sample' forecast. ie, given x_{n+1} and $\hat{\beta}$,

$$\hat{y}_{n+1} = x_{n+1}' \hat{\beta}.$$

- The forecast error: $\hat{e}_{n+1} = y_{n+1} - x_{n+1}' \hat{\beta}$ has MSE

$$E[\hat{e}_{n+1}^2] = E[e_{n+1}^2] - 2E[e_{n+1} x_{n+1}' (\hat{\beta} - \beta)] + E[x_{n+1}' V_{\hat{\beta}} x_{n+1}].$$

Assuming homoskedasticity and no serial correlation,

$$E[\hat{e}_{n+1}^2] = \sigma^2 \left(1 + E[x_{n+1}' (X'X)^{-1} x_{n+1}] \right)$$

Omitted variables bias

Let $X = [X_1, X_2]$. Suppose the correct model is

$$y = X_1\beta_1 + X_2\beta_2 + e = X\beta + e$$

A regression of y on X_1 yields

$$\begin{aligned}\hat{\beta} &= (X_1'X_1)^{-1}X_1'y \\ &= (X_1'X_1)^{-1}X_1'(X_1\beta_1 + X_2\beta_2 + e) \\ &= \beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2 + (X_1'X_1)^{-1}X_1'e\end{aligned}$$

$E(\hat{\beta}_1|X) \neq \beta_1$ unless $X_1'X_2 = 0$.

Inclusion of Irrelevant Regressors

Now suppose the true model is $y = X_1\beta_1 + e$ and we estimate a model including X_2 as regressors.

- True model is $y = X_1\beta_1 + X_2\beta_2 + e$ with $\beta_2 = 0$
- $E(\hat{\beta}|X) = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} \beta_1 \\ 0 \end{pmatrix}.$
- the cost in including too many regressors is efficiency because the covariance matrix without X_2 is never larger than the one with X_2 .

Mean squared error trade-off

Suppose true model is $y = X_1\beta_1 + X_2\beta_2 + e$

- $\text{var}(\hat{\beta}_1|X_1, X_2) = \sigma^2(X_1'X_1 - X_1'X_2(X_2'X_2)^{-1}X_2'X_1)^{-1}$
- if X_2 is dropped, $\hat{\beta}_1$ is biased but $\text{var}(\hat{\beta}_1|X_1) = \sigma^2(X_1'X_1)^{-1}$.
- It can be shown that

$$\text{var}(\hat{\beta}_1|X_1)^{-1} - \text{var}(\hat{\beta}_1|X_1, X_2)^{-1} = \sigma^2(X_1'X_2(X_2'X_2)^{-1}X_2'X_1) > 0$$

- although $\hat{\beta}_1$ is biased, it may mean square dominate the estimate from the long regression.

Ridge Estimation

- $\text{svd}(X) = UDV'$: $X'X = VD^2V'$: $D_{n \times K}$ is diag, $U_{n \times n}$ and $V_{K \times K}$ are orthonormal.

$$\hat{\beta} = (VD^2V')^{-1}VDU'y.$$

- OLS unreliable when columns of X are collinear, especially when K is large. Add λ to diagonal reconditions $X'X$.

- Ridge estimator: $\hat{\beta}_R = (X'X + \lambda I_K)^{-1}X'y$

$$\begin{aligned}\hat{\beta}_R &= (VD^2V' + \lambda VV')^{-1}VDU'y \\ &= V(D^2 + \lambda I_K)^{-1}V'VDU'y \\ &= V(D^2 + \lambda I_K)^{-1}DU'y.\end{aligned}$$

- biased (because the unconstrained estimator is unbiased), but if $0 < \lambda < 2\sigma^2/\beta'\beta$, $\text{mse}(\hat{\beta}_R|X) < \text{mse}(\hat{\beta}|X)$.

Ridge Prediction

$$\begin{aligned}
 \hat{y}_R &= X\hat{\beta}_R = UD(D^2 + \lambda I)^{-1}DU'y \\
 &= \sum_{j=1}^K U_j \left(\frac{d_j^2}{d_j^2 + \lambda} \right) U_j'y \\
 \hat{y}_{OLS} &= X\hat{\beta} = \sum_{i=1}^K U_i U_i'y.
 \end{aligned}$$

- ridge fit projects y onto components with large d_j .
- a small d_j are directions in column space of X with small variance. Ridge regressions shrink these most.
- The degrees of freedom is

$$\text{df}(\lambda) = \text{tr}(X(X'X + \lambda I)^{-1}X') = \sum_{j=1}^K \frac{d_j^2}{d_j^2 + \lambda}.$$

The Classical Normal Linear Model

If $e \sim N(0, D)$ and the regressors X are nonstochastic,

$$\begin{aligned}\hat{\beta} - \beta &= (X'X)^{-1}X'e \\ &= (X'X)^{-1}X' \times N(0, D) \\ &\sim N(0, (X'X)^{-1}X'DX(X'X)^{-1})\end{aligned}$$

since $z \sim N(\mu, D) \Rightarrow Az + b \sim N(A\mu + b, ADA')$.

- we will use asymptotic theory since this allows e to be nonnormal. But this requires n to be large.

Convergence in Probability

- A random sequence $b_1, b_2, \dots$, converges in probability to the limit b if for every $\epsilon > 0$, $\lim_{n \rightarrow \infty} P(|b_n - b| \geq \epsilon) = 0$. This is also written $\text{plim } b_n = b$.
- Mean-squared convergence implies convergence in probability.

Proof: by Markov inequality, for any $\epsilon > 0$,

$$\begin{aligned} P(|b_n - b| > \epsilon) &= P((b_n - b)^2 > \epsilon^2) \\ &\leq \mathbb{E}[(b_n - b)^2] / \epsilon^2 = \text{var}(b_n) / \epsilon^2 \rightarrow 0. \end{aligned}$$

$b_n \xrightarrow{m.s.} b \Rightarrow \lim_{n \rightarrow \infty} \mathbb{E}[(b_n - b)^2] = 0$. Thus, $b_n \xrightarrow{p} b$.

- An estimator $\hat{\beta}$ is consistent for β if $\text{plim } \hat{\beta} = \beta$.

Law of Large Numbers

- Use LLN to get probability limit when $b_n = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.
- If a LLN can be applied, then $\bar{X}_n - \mu \xrightarrow{p} 0$.

$$\text{plim } \bar{X}_n = \lim \mathbb{E}[\bar{X}_n] = \lim \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mu.$$

- Khinchine's WLLN: Let $\{X_i\}$ be iid and $\mathbb{E}[X_i]$ exists.
- Kolmogorov's LLN for inid data
- Ergodic Theorem for dependent data.

Consistency of Sample Mean

Let $X_1, \dots, X_n$ be iid samples with mean μ . Let $b_n = \bar{X}_n$. Since $\mathbb{E}[\bar{X}] = \mu$ by linearity of expectation,

$$\begin{aligned} E[(\bar{X}_n - \mu)^2] &= \text{var}(\bar{X}_n) = E\left[\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu\right)^2\right] \\ &= \text{var}\left(\frac{1}{n} \sum_i X_i\right) = \frac{1}{n^2} \text{var}\left(\sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \left(\sum_{i=1}^n \text{var}(X_i)\right) \text{ since } X_i \text{ are iid} \\ &= \frac{n \cdot \text{var}(X)}{n^2} = \frac{\text{var}(X)}{n} \rightarrow 0 \text{ as } n \rightarrow \infty. \end{aligned}$$

Hence $\bar{X}_n$ converges in mean-square to μ .